

18:44pm
02-07-2026



k-Nearest Neighbors

$x \rightarrow 1$
 $x \rightarrow 0$

$k=3$ means 3 nearest neighbors.

Euclidean distance

100 distances

sort ascending closest

majority count

1 1 0 $\therefore$ majority = 1

so class will be 1

```
df = pd.read_csv('/kaggle/input/breast-cancer-wisconsin/data/data.csv')
df.head()
```

```
df.drop(columns=['id', 'Unnamed: 32'], inplace=True)
```

```
df.head()
```

→ Removing columns

```
df.shape
```

```
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(df.iloc[:, 1:],
                                                    df.iloc[:, 0], test_size=0.2, random_state=2)
```

```
X_train.head()
```

```
X_train.shape
```

```
from sklearn.preprocessing import StandardScaler
scaler = StandardScaler()
```

```
X_train = scaler.fit_transform(X_train)
```

```
X_test = scaler.transform(X_test)
```

```
X_train
```

→ get numpy array

```
X_train.shape
```

```
from sklearn.neighbors import KNeighborsClassifier
```

```
knn = KNeighborsClassifier(n_neighbors=5)
```

```
knn.fit(X_train, y_train)
```

KNN rely on distance

So, always use transformation before applying KNN

StandardScaler -

for each group =

$$\frac{x_i - \bar{x}}{\sigma}$$

```

from sklearn.metrics import accuracy_score
y_pred = knn.predict(X_test)
accuracy_score(y_test, y_pred)

```

$\hookrightarrow 0.973$
 Scores = [] # Cross-validation:

for i in range(1, 16):

knn = KNeighborsClassifier(n_neighbors = i)

knn.fit(X_train, y_train)

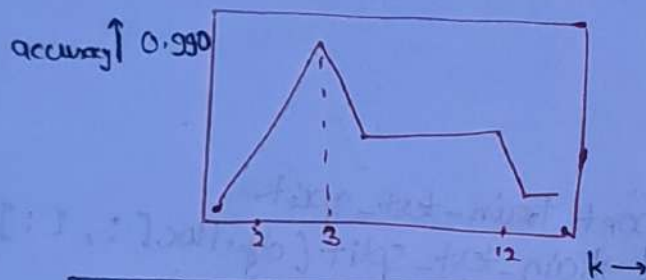
y_pred = knn.predict(X_test)

scores.append(accuracy_score(y_test, y_pred))

```

import matplotlib.pyplot as plt
plt.plot(range(1, 16), scores)

```



at $k=3$, we will get
accuracy = 0.990

cgpa	iq	Placement	Prediction
8	80	1	1 ✓
9	90	0	1 x (wrong prediction)

$$\text{Accuracy-score} = \frac{\text{Right Prediction}}{\text{Total Students}}$$

$$= \frac{100}{200} = 50\% \text{ accuracy}$$

200 rows

100 ✓

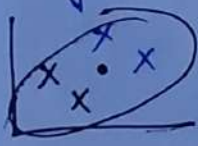
How to Select k?

Heuristic Techni.
(Jugaad)

$\sqrt{n}$ $n \rightarrow$ observ.

$$400 \rightarrow \sqrt{400} = 20$$

* Avoid to take even value
for k.



2(1) - 2(0)

conflicts

Experimentation

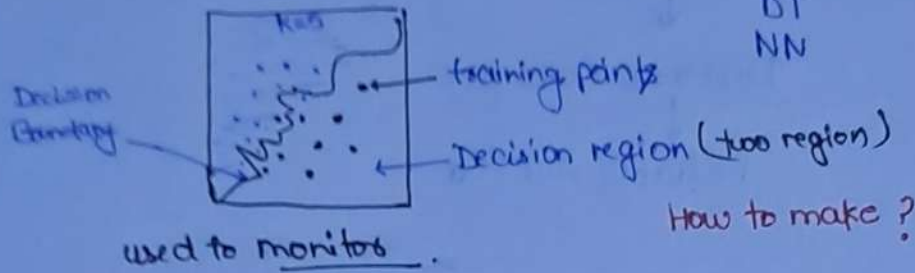
• Cross-validation

diff knn $\begin{cases} k=1 \\ k=2 \\ k=3 \\ k=4 \\ \vdots \end{cases}$

* then find accuracy score.
and select the highest accuracy score among them

Decision Surface/Boundary : is a tool used for Classification problems

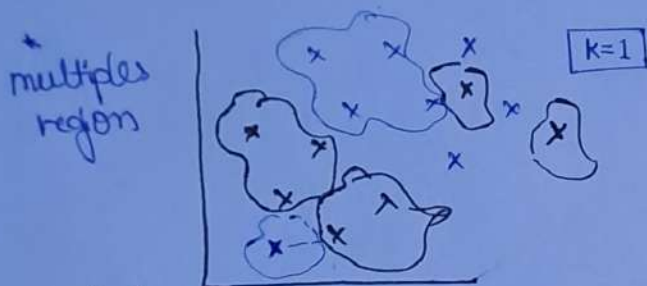
like :
SVM
LR
DT
NN



mlxtend Library
↳ To plot decision Tree

How to make ?

Overfitting and Underfitting in KNN :



minor changes
on data set
↓
Overfitting
↓
High Variance

Q. what will be the max value of K ?

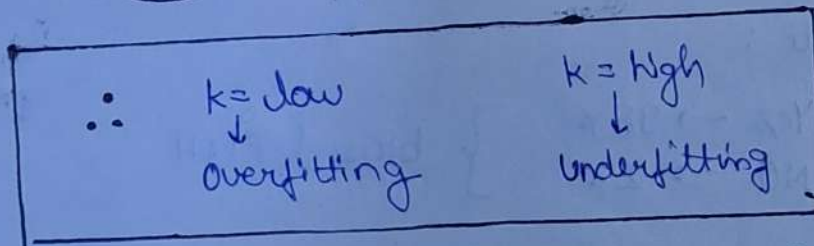
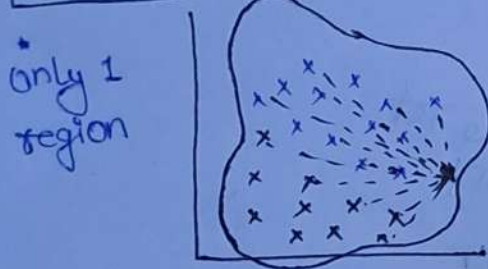
Ans. $K = \text{no. of Observations}$

$K=200$ → simplistic → Underfitting

Blue > pink

then, Blue always will in majority counts,

Labels
x → Blue
x → Pink



Limitations of KNN : How and when to use knn ?

failure case :

① Large Datasets ($n=5L, f=100$)
knn is a lazy learning technique.

Training phase : you actually do nothing, (points stores)

Testing phase : query point → fig(b) at training time → prediction

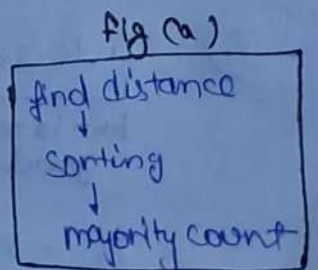
Result: Too slow

Ideal case:

Low Latency

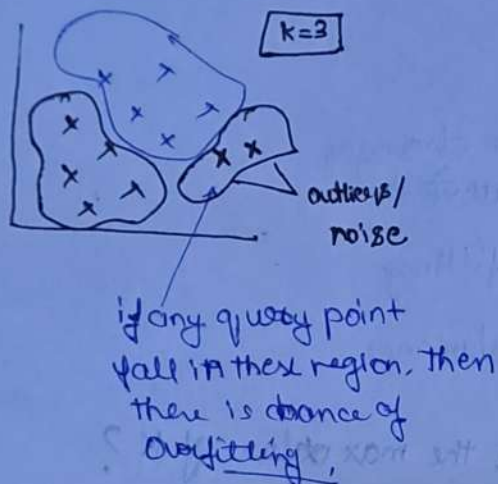
∴ Highly Computation time
for finding Distance and
all.

Latency ↑↑



- ② High dimensional data : f. features = 500
 ↓
 Curse of dimensionality
 ↓
 Distance Concept
 ↓
 Reliable नहीं होता

③ Outliers :



④ Non-Homogeneous Scalars :

experience	salary	fine
0-25	20k - 1Gr	

Distances reliable नहीं होता

⑤ Imbalanced Datasets :

Classes के बीच : Yes → 98%
 NO → 2% } biased होता

⑥ Inference and Not for prediction :
 knn is not good. knn is good

- x_1, x_2 का impact on y
- which feature is more important
- Black-box model